

# CloudFormation

Class no:26

## What is AWS CloudFormation?

- **Definition:** CloudFormation is an **Infrastructure-as-Code (IaC)** service that lets you **model, provision, and manage AWS resources** using templates.
- **Purpose:** Automate and standardize the creation, updating, and deletion of AWS resources (EC2, S3, RDS, VPC, etc.) in a repeatable, version-controlled way.

## Core concepts

- **Template**
  - A text file in **JSON or YAML** format that describes the AWS resources and their properties.
  - Acts as a **blueprint** for your infrastructure (what to create and how it should be configured).
- **Stack**
  - A **collection of AWS resources** created from a single template.
  - You can **create, update, or delete** a stack, and CloudFormation manages all underlying resources as a single unit.

## Basic template structure (YAML example)

- **AWSTemplateFormatVersion:** Version of template syntax.
- **Parameters:** Inputs you pass when creating/updating the stack (e.g., environment, instance type).
- **Resources:** The actual AWS resources (EC2, S3, etc.) and their properties.
- **Outputs:** Values you can reference later (e.g., bucket URL, instance IP).

## Key features and benefits

- **Infrastructure as Code:** Store templates in **Git / S3**, enabling version control, code reviews, and CI/CD.
- **Repeatable deployments:** Same template always creates the **same infrastructure** in dev, test, and prod (with parameters).
- **Dependency management:** CloudFormation automatically handles **resource ordering and dependencies** (e.g., VPC before subnets).
- **Rollback and change management:** If a stack update fails, CloudFormation can **roll back** to the previous working state. You can use **change sets** to review changes before applying them.

## **Hands on: To create an EC2 template using Cloud formation:**

**Step no:1** Create a **yaml template file** for EC2 and store it as .yaml file.

**Code for reference:** I saved it as **AWSTemplateFormatVersion.yaml**

AWSTemplateFormatVersion: "2010-09-09"

Description: |

Creates a single EC2 instance with a security group allowing SSH.

Parameters:

KeyName:

Type: AWS::EC2::KeyPair::KeyName

Description: Name of an existing EC2 KeyPair to enable SSH access.

InstanceType:

Type: String

Default: t3.micro

AllowedValues:

- t3.micro

- t3.small

Description: Instance type for the EC2 instance.

SSHLocation:

Type: String

Default: 0.0.0.0/0

Description: IP range allowed to SSH to the instance.

Resources:

EC2InstanceSecurityGroup:

Type: AWS::EC2::SecurityGroup

Properties:

GroupDescription: Enable SSH access

SecurityGroupIngress:

- IpProtocol: tcp

FromPort: 22

ToPort: 22

CidrIp: !Ref SSHLocation

EC2Instance:

Type: AWS::EC2::Instance

Properties:

ImageId: ami-0e12ffc2dd465f6e4 # <-- Replace with a valid AMI in your region

InstanceType: !Ref InstanceType

KeyName: !Ref KeyName

SecurityGroupIds:

- !Ref EC2InstanceSecurityGroup

Tags:

- Key: Name

Value: MyCloudFormationEC2Instance

Outputs:

InstanceId:

Description: Instance ID of the EC2 instance

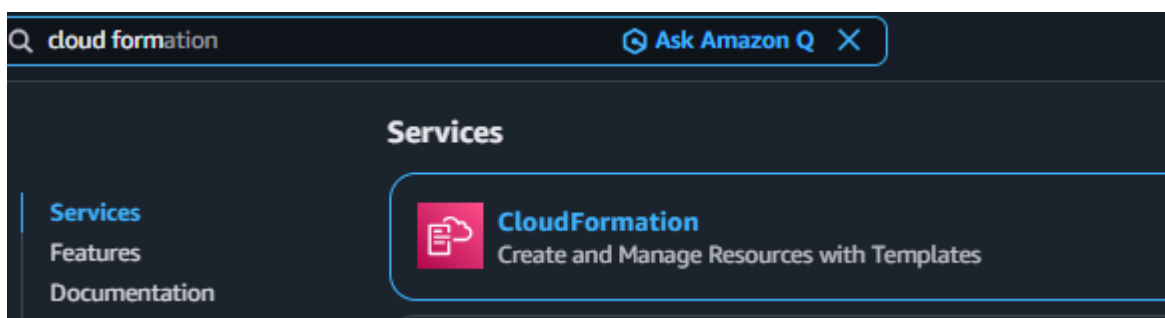
Value: !Ref EC2Instance

PublicIp:

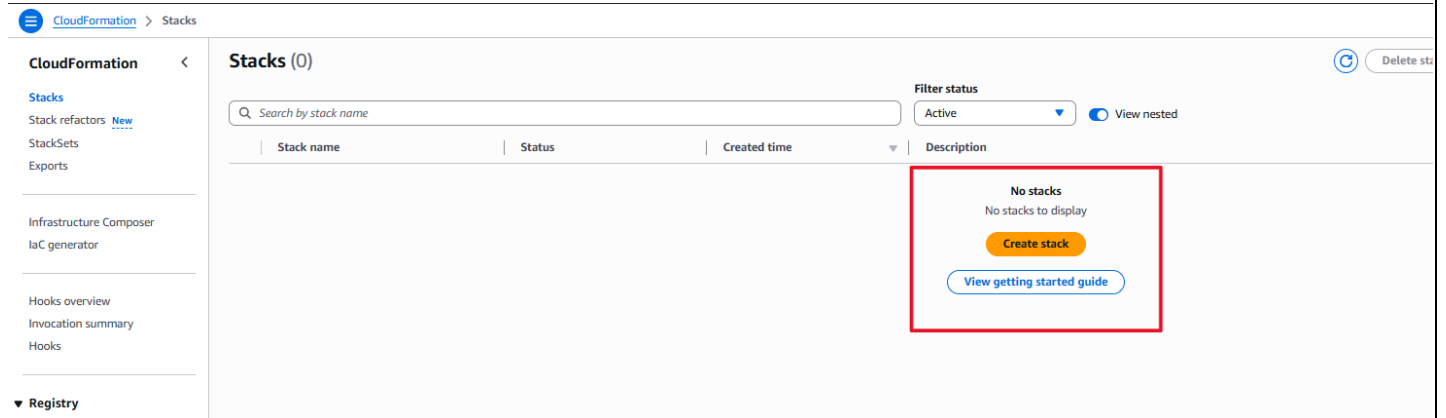
Description: Public IP address of the EC2 instance

Value: !GetAtt EC2Instance.PublicIp

**Step no:2** Go to the AWS console → Search **CloudFormation** → click open.



## Create Stack.



Choose: **Existing template**

### Upload a template file

Choose our file and click **upload**

**Note** the file will be uploaded. AWS will create a bucket and it is saved in it. As we can see the **S3 url** once we have uploaded our file.

### Create stack

#### Prerequisite - Prepare template

You can also create a template by scanning your existing resources in the [IaC generator](#).

#### Prepare template

Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

##### ☒ Choose an existing template

Upload or choose an existing template.

##### ☐ Build from Infrastructure Composer

Create a template using a visual builder.

#### Specify template [Info](#)

This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

#### Template source

Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

##### ☐ Amazon S3 URL

Provide an Amazon S3 URL to your template.

##### ☒ Upload a template file

Upload your template directly to the console.

##### ☐ Sync from Git

Sync a template from your Git repository.

#### Upload a template file

##### [Choose file](#)

AWSTemplateFormatVersion.yaml

JSON or YAML formatted file

S3 URL: <https://s3.ap-south-1.amazonaws.com/cf-templates-1wli9mrarvbe-ap-south-1/2026-04-17T120741.681Z2yr-AWSTemplateFormatVersion.yaml>

[View in Infrastructure Composer](#)

[Cancel](#)

[Next](#)

Provide the stack name: **myec2stack**

The **parameters** for the EC2 is read from the yaml file and will be auto populated here.

### Specify stack details

#### Provide a stack name

Stack name

myec2stack

Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 10/128.

#### Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

##### InstanceType

Instance type for the EC2 instance.

t3.micro

##### KeyName

Name of an existing EC2 KeyPair to enable SSH access.

KeyName

##### SSHLocation

IP range allowed to SSH to the instance.

0.0.0.0/0

Cancel

Previous

Next

Keep the rest as default.

### Configure stack options

#### Tags - optional

Tags (key-value pairs) are used to apply metadata to AWS resources, which can help in organizing, identifying, and categorizing those resources. You can add up to 50 unique tags for each stack.

No tags associated with the stack.

Add new tag

You can add 50 more tag(s)

#### Permissions - optional

Specify an existing AWS Identity and Access Management (IAM) service role that CloudFormation can assume.

##### IAM role - optional

Choose the IAM role for CloudFormation to use for all operations performed on the stack.

IAM role name

Sample-role-name

Remove

Create new role



#### Stack failure options

##### Behavior on provisioning failure

Specify the roll back behavior for a stack failure. [Learn more](#)

###### ☒ Roll back all stack resources

Roll back the stack to the last known stable state.

###### ☐ Preserve successfully provisioned resources

Preserves the state of successfully provisioned resources, while rolling back failed resources to the last known stable state. Resources without a last known stable state will be deleted upon the next stack operation.

##### Delete newly created resources during a rollback

Specify whether resources that were created during a failed operation should be deleted regardless of their deletion policy. [Learn more](#)

###### ☒ Use deletion policy

Retains or deletes created resources according to their attached deletion policy.

###### ☐ Delete all newly created resources

Deletes created resources during a rollback regardless of their attached deletion policy.

Next → Next.

Now **Review and submit.**

## Review and create

### Step 1: Specify template

Edit

#### Prerequisite - Prepare template

**Template**  
Template is ready

#### Template

**Template URL**  
<https://s3.ap-south-1.amazonaws.com/cf-templates-1wii9mrarvbe-ap-south-1/2026-04-17T123855.822Zotq-AWSTemplateFormatVersion1.yaml>

**Stack description**  
Creates a single EC2 instance with a security group allowing SSH.

### Step 2: Specify stack details

Edit

#### Provide a stack name

**Stack name**  
myec2stack

#### Parameters (3)

| Q Search parameters |  | < 1 > ⚙   |  |
|---------------------|--|-----------|--|
| Key                 |  | Value     |  |
| InstanceType        |  | t3.micro  |  |
| KeyName             |  | KeyName   |  |
| SSHLocation         |  | 0.0.0.0/0 |  |

### Step 3: Configure stack options

Edit

#### Rollback configuration

**Monitoring time**  
-

**CloudWatch alarm ARN**  
-

#### Notification options

**SNS topic ARN**

No notification options  
There are no notification options defined

#### Stack creation options

**Timeout**  
-

**Termination protection**  
Deactivated

#### Quick-create link

Use quick-create links to get stacks up and running quickly from the AWS CloudFormation console with the same basic configuration as this stack. Copy the URL on the link to share. [Learn more](#)

[Open quick-create link](#)

[Create change set](#)

[Cancel](#)

[Previous](#)

[Submit](#)

**Step no:3** If there are no issues in the yaml file a stack will be created. The **status become: CREATE\_COMPLETE**.

Stacks (1)

Search by stack name

Filter status: Active View nested

| Stack name | Status          | Created time                 | Description                                                       |
|------------|-----------------|------------------------------|-------------------------------------------------------------------|
| myec2stack | CREATE_COMPLETE | 2026-04-17 18:10:19 UTC+0530 | Creates a single EC2 instance with a security group allowing SSH. |

Go to the **EC2 console** → we can see an instance has been created successfully using the cloud formation.

Instances (1) Info

Last updated less than a minute ago

Saved filter sets: Choose filter set

Find Instance by attribute or tag (case-sensitive)

| Name                        | Instance ID         | Instance state | Instance type |
|-----------------------------|---------------------|----------------|---------------|
| MyCloudFormationEC2Instance | i-037cedbb94c0c73bf | Running        | t3.micro      |

**Step no:4** Now change the name of the instance from **MyCloudFormationEC2Instance** (provided in the template) to **Myinstance**. We can see the drift changes in the cloud formation.

Instances (1/1) Info

Last updated less than a minute ago

Saved filter sets: Choose filter set

Find Instance by attribute or tag (case-sensitive)

| Name       | Instance ID         | Instance state | Instance type |
|------------|---------------------|----------------|---------------|
| Myinstance | i-037cedbb94c0c73bf | Running        | t3.micro      |

Now go to **cloud formation** → **select our stack** → **actions** → **detect drifts**

Stacks (1)

Search by stack name

Filter status: Active View nested

Stack actions: Delete stack Update stack Stack actions Create stack

| Stack name | Status          | Created time                 | Description                                                       |
|------------|-----------------|------------------------------|-------------------------------------------------------------------|
| myec2stack | CREATE_COMPLETE | 2026-04-17 18:10:19 UTC+0530 | Creates a single EC2 instance with a security group allowing SSH. |

- Edit termination protection
- View drift results
- Detect drift
- Create a change set
- Import resources into stack
- Cancel update stack
- Continue update rollback
- View in Application Manager

Click **Detect stack drifts** and refresh.

Stack drift status

Drift detection enables you to detect whether a stack's actual configuration differs, or has drifted, from its template configuration. [Learn more](#)

Drift status

⚠️ DRIFTED

Last drift check time

2026-04-17 18:17:41 UTC+0530

ⓘ This stack contains drifted resources. Create and execute a drift-aware change set to bring all stack resources in line with the template from the previous deployment.

Create a change set

Resource drift status (2)

Only resources which currently support drift detection are displayed here. To view all of your stack resources, see your stack details page. [Learn more](#)

Q Search resources

< 1 >

| Logical ID               | Physical ID                                                      | Type                    | Drift status | Timestamp                    | Module | Drift status reason |
|--------------------------|------------------------------------------------------------------|-------------------------|--------------|------------------------------|--------|---------------------|
| EC2InstanceSecurityGroup | <a href="#">myec2stack-EC2InstanceSecurityGroup-vbkVUrPdLnIt</a> | AWS::EC2::SecurityGroup | 🟢 IN_SYNC    | 2026-04-17 18:17:42 UTC+0530 | -      | -                   |
| EC2Instance              | <a href="#">i-037cedbb94c0c73bf</a>                              | AWS::EC2::Instance      | ⚠️ MODIFIED  | 2026-04-17 18:17:42 UTC+0530 | -      | -                   |

Now select the **Modified resource** → we can compare the expected template parameters and the actual parameters.

As we changed the name of the instance we can see that here.

Differences (2)

☐ Property

▲ Change

▼ Expected value

▼ Current value

|                                             |           |                                                  |                      |
|---------------------------------------------|-----------|--------------------------------------------------|----------------------|
| <input type="checkbox"/> SecurityGroupIds.0 | NOT_EQUAL | myec2stack-EC2InstanceSecurityGroup-vbkVUrPdLnIt | sg-0b84da4f46dcd36a1 |
| <input type="checkbox"/> Tags.0.Value       | NOT_EQUAL | MyCloudFormationEC2Instance                      | Myinstance           |

Details

Expected

Actual

```
{
  "KeyName": "KeyName",
  "ImageId": "ami-0e12ffc2dd465f6e4",
  "InstanceType": "t3.micro",
  "SecurityGroupIds": [
    "myec2stack-EC2InstanceSecurityGroup-vbkVUrPdLnIt"
  ],
  "Tags": [
    {
      "Value": "MyCloudFormationEC2Instance",
      "Key": "Name"
    }
  ]
}
```

```
{
  "ImageId": "ami-0e12ffc2dd465f6e4",
  "KeyName": "KeyName",
  "InstanceType": "t3.micro",
  "SecurityGroupIds": [
    "sg-0b84da4f46dcd36a1"
  ],
  "Tags": [
    {
      "Value": "Myinstance",
      "Key": "Name"
    }
  ]
}
```



## Changing the type of an already existing instance:

**Step no:1** Create and launch an instance.

Once initialized → **instance state** → stop.

Instances (1/1) Info

Connect Instance state Actions

Find Instance by attribute or tag (case-sensitive)

| Name       | Instance ID         | Instance state | Instance type | Status check | Alarm status  | Availability |
|------------|---------------------|----------------|---------------|--------------|---------------|--------------|
| Myinstance | i-037cedbb94c0c73bf | Stopping       | t3.micro      | -            | View alarms + | ap-south-1   |

### i-037cedbb94c0c73bf (Myinstance)

| Details                                                                | Status and alarms | Monitoring                                                                      | Security | Networking                              | Storage | Tags |
|------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------|----------|-----------------------------------------|---------|------|
| ▼ Instance summary Info                                                |                   |                                                                                 |          |                                         |         |      |
| Instance ID<br>i-037cedbb94c0c73bf                                     |                   | Public IPv4 address<br>-                                                        |          | Private IPv4 addresses<br>172.31.46.188 |         |      |
| IPv6 address<br>-                                                      |                   | Instance state<br>Stopped                                                       |          | Public DNS<br>-                         |         |      |
| Hostname type<br>IP name: ip-172-31-46-188.ap-south-1.compute.internal |                   | Private IP DNS name (IPv4 only)<br>ip-172-31-46-188.ap-south-1.compute.internal |          |                                         |         |      |

**Step no:2** Once stopped → click actions → **Instance settings** → change instance type

Instances (1/1) Info

Connect Instance state Actions Launch instance

Find Instance by attribute or tag (case-sensitive)

| Name       | Instance ID         | Instance state | Instance type |
|------------|---------------------|----------------|---------------|
| Myinstance | i-037cedbb94c0c73bf | Stopped        | t3.micro      |

Instance diagnostics

Instance settings

Networking

Security

Image and templates

Storage

Monitor and troubleshoot

Protection and lifecycle management

Attach to Auto Scaling Group

Change stop protection

Change termination protection

Change shutdown behavior

Change auto-recovery behavior

Change instance migration on reboot

Protect with Elastic Disaster Recovery

Instance configuration

Change instance type

Change CPU options

Modify instance placement

Modify Capacity Reservation settings

Change Nitro Enclaves

Connect RDS database

Change credit specification

Change resource based naming options

Edit user data

### i-037cedbb94c0c73bf (Myinstance)

| Details                                                                | Status and alarms | Monitoring                                                                      | Security | Networking | Storage | Tags |
|------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------|----------|------------|---------|------|
| ▼ Instance summary Info                                                |                   |                                                                                 |          |            |         |      |
| Instance ID<br>i-037cedbb94c0c73bf                                     |                   | Public IPv4 address<br>-                                                        |          |            |         |      |
| IPv6 address<br>-                                                      |                   | Instance state<br>Stopped                                                       |          |            |         |      |
| Hostname type<br>IP name: ip-172-31-46-188.ap-south-1.compute.internal |                   | Private IP DNS name (IPv4 only)<br>ip-172-31-46-188.ap-south-1.compute.internal |          |            |         |      |

Choose the desired type → Click change instance type.

Change instance type

[Info](#) | [Get advice](#)

You can change the instance type only if the current instance type and the instance type that you want are compatible.

Instance ID

i-037cedbb94c0c73bf

(MyInstance)

Current instance type

t3.micro

New instance type

t3.micro

Change instance type to: t3.micro

c7i-flex.large

m7i-flex.large

t3.micro

t3.small

(32-bit) / (64-bit) architecture can be chosen.

is instance type

|                             | t3.micro            |
|-----------------------------|---------------------|
| On-Demand Linux pricing     | 0.0112 USD per Hour |
| On-Demand Windows pricing   | 0.0204 USD per Hour |
| vCPUs                       | 2 (1 core)          |
| Memory (MiB)                | 1024                |
| Storage (GB)                | —                   |
| Supported root device types | ebs                 |
| Network performance         | Up to 5 Gigabit     |
| Architecture                | x86_64              |
| Hypervisor                  | nitro               |
| Burstable                   | true                |
| Free-tier eligible          | true                |
| Current generation          | true                |

[Compare more instance type attributes](#)

Advanced details

CPU options - update

[Info](#)

☒ Use default CPU options

☐ Specify CPU options

Configure CPUs for your instance to optimize performance and save on licensing costs.

Active vCPUs

2

Total vCPUs

2

Cancel

Change instance type